

EMC ASSESSMENT (EU – SWITCHING POWER SUPPLY DEVICE)

Product: SidecarTridge USB-C PD PSU

Model/Revision: SidecarTridge USB-C PD Multi-Rail PSU for Atari ST (Rev.1.1.0) / REV1.1.0

1. PRODUCT OVERVIEW

The product is a low-voltage DC-DC power conversion device intended for use in retro computing systems. It converts externally supplied low-voltage DC input (e.g. via USB-C Power Delivery) into multiple regulated DC output rails required by the host system.

The device may be used as an internal replacement power supply or as an external power adapter module depending on configuration.

Typical operating voltage is below 50 V DC.

2. APPLICABLE DIRECTIVE

This assessment is made in relation to:

- Directive 2014/30/EU (Electromagnetic Compatibility – EMC)

The product does not fall under Directive 2014/35/EU (Low Voltage Directive) as it operates below 50 V DC and is not directly connected to mains supply.

The external AC/DC adapter (e.g. USB-C PD power supply) is outside the scope of this assessment and must itself comply with applicable safety and EMC legislation.

3. DESIGN CHARACTERISTICS RELEVANT TO EMC

- The device incorporates DC-DC switching regulators to generate multiple regulated output voltages.
- Switching frequencies are typically in the range of hundreds of kHz to low MHz.
- The device includes control circuitry, power inductors, switching elements, and output filtering components.
- Appropriate PCB layout, grounding, decoupling, and filtering techniques have been implemented to limit conducted and radiated emissions.
- No radio transmitter or intentional RF radiator is present.
- The device does not include wireless communication functionality.
- External connections include power input (e.g. USB-C) and low-voltage output rails to the host system.
- External cables (input and output) may act as potential emission paths.
- The device may be powered via a USB-C connector used solely for low-voltage DC power input; no USB data interface is implemented.

4. EMISSION ASSESSMENT

Due to the presence of switching regulators:

- The device inherently generates high-frequency switching noise as part of normal operation.
- Emissions are mitigated through circuit design techniques including filtering, decoupling, and controlled switching behavior.
- The relatively low power level and short cable lengths typical of the intended use reduce the likelihood of significant radiated emissions.
- The absence of mains connection and the use of an external compliant AC/DC adapter limit conducted emissions to low-voltage domains.

Conclusion:

The product presents a controlled and acceptable level of electromagnetic emissions when used as intended and in conjunction with compliant external power supplies.

5. IMMUNITY ASSESSMENT

- The device is designed to operate in typical residential, commercial, and light-industrial electromagnetic environments.
- Immunity performance depends on the quality and stability of the external DC supply (e.g. USB-C PD adapter).
- The device includes standard protection features such as input filtering, voltage regulation, and protection circuitry (e.g. overcurrent and thermal protection).
- Output stability is maintained through regulated DC-DC conversion and filtering.
- The design includes protection features such as overcurrent and thermal shutdown to ensure safe operation under fault conditions.

Conclusion:

The product is expected to operate correctly in typical electromagnetic environments when used with compliant external power supplies and appropriate cabling.

6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into or connection with compatible host systems.
 - It may be used either as an internal replacement PSU or as an external power adapter depending on configuration.
 - Proper EMC performance depends on correct installation, cable routing, and use of a compliant external power supply.
 - The external AC/DC adapter (USB-C PD supply) is not part of this product and must be CE-marked.
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7. OVERALL CONCLUSION

Based on the design characteristics and intended use, the product is considered to comply with the essential requirements of Directive 2014/30/EU (EMC), provided that it is installed and used in accordance with its intended purpose.

This assessment is supported by the use of harmonised standards such as:

- EN IEC 61000-6-1:2019 (Immunity)
- EN IEC 61000-6-3:2021 (Emissions)

No additional EMC mitigation measures are required beyond standard design practices for this class of device.

8. RESPONSIBLE PERSON

Name: Diego Parrilla Santamaria
Position: CEO
Company: GOODDATA LABS SL

Date: 23 April 2026

Signature:

GOODDATA LABS SL
CIF: B09834704
C/BERLIN 16, BLD. 2ºA
28022 - ALCAZOVÉN (MADRID)

